

Euler-Lagrange Modeling and Control of Quadrotor UAV with Aerodynamic Compensation

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Abstract—Research focuses on deriving an accurate mathematical model for quadrotor UAVs based on the Euler-Lagrange formulation, which also includes gyroscopic effects and aerodynamic drag. The obtained model is implemented in Matlab/Simulink and compared against a Simscape Multibody implementation. Then, a PD hierarchical control with exact linearization is implemented, tested and compared with the same controller but without considering aerodynamic effects. Results show almost identical behavior, but performance is slightly better when full dynamic compensation is followed. Both detailed models are also released and available as Open Source to be used for controller design, testing and evaluation.

Index Terms—Quadrotor, Modelling, Control, Simscape

I. INTRODUCTION

Unmanned aerial vehicles (UAVs) are widely used in a plethora of military, civilian and public domain applications [1]. Among UAVs, quadrotors and rotorcrafts attract much attention because of their ability to perform vertical take off and landing (VTOL) and hovering, which makes them attractive for diverse applications [2] like wind turbine inspection [3], bridge inspection [4] and fire monitoring [5]. Focusing on the conventional quadrotor configuration (with different propeller rotation speed and direction that allows for motion in 3D space [6]), although they have a relatively simple mechanical structure, their highly nonlinear and coupled multiple-input multiple-output dynamics present challenges when it comes to controller design. These challenges are more difficult to overcome when quadrotor model parametric and unmodeled uncertainties and external disturbances are considered [2]. Moreover, quadrotors are underactuated systems featuring four motors to control six generalized coordinates (position and attitude); this basically dictates the need to design and implement a hierarchical control

architecture for trajectory tracking [7].

Quadrotor UAV mathematical modeling mostly follows either the Newton Euler [8, 9, 10] or the Euler Lagrange formulation [6, 11, 12]. The main difference when deriving the equations of motion is that in the Newton Euler formulation, motion is described in terms of force and moment conservation, while the Euler Lagrange formulation follows derivation of a scalar energy-based function with respect to a set of generalized coordinates [13]. Regardless, reported research shows that quadrotor modeling also includes models based on static thrust characteristics [6, 7, 10, 11] that is reliable under hovering conditions but is limited in accuracy during motion due to aerodynamic drag. In [12] a linear model is used to account for aerodynamic effects, while in [9] a nonlinear model for the drag force is adopted. Both approaches do not consider the different types of drag acting on the quadrotor. In [14], a detailed aerodynamic model is introduced to reduce modeling errors during aggressive manoeuvres. The model of [14] is derived using the Newton Euler formulation, but, to the best of the authors knowledge, no quadrotor model derived from the Lagrange formulation with the aerodynamic effects presented in [14] has been implemented.

The aim of this research is to provide a free access Matlab/Simulink and Simscape Multibody quadrotor model with aerodynamic effects to be used for controller design, implementation and testing. The Simulink model is implemented using the Euler Lagrange formulation since it is easy and systematic to derive, and it allows for a compact analytical matrix form, which is advantageous when designing controllers [13]. Then, a PD hierarchical control with exact linearization is implemented, tested and compared with the same controller but without considering aerodynamic effects.

The rest of this paper is organised as follows: Section II describes the Lagrange modeling of the quadrotor that includes aerodynamic and gyroscopic effects. Section III introduces the Simulink and Simscape model schematic and performs a comparison between the two models. In Section IV the control problem is considered and in Section V a hierarchical PD controller with exact linearization is simulated for tracking a helix trajectory and it is compared to a controller without considering aerodynamic effects. Lastly, in Section VI, conclusions and future development are discussed.

II. LAGRANGE MODELLING OF QUADROTOR UAV

The quadrotor under study is shown in Fig. 1. The quadrotor body is considered rigid and symmetric, with the arms aligned to x_b and y_b axes [10].

Due to the symmetric structure, the inertia matrix I_B in the

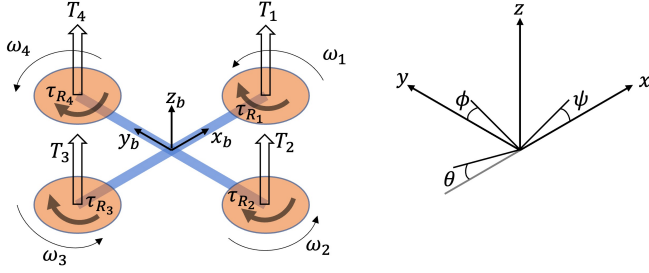


Fig. 1. Quadrotor inertial and body frame.

body frame is diagonal and $I_x = I_y$.

$$I_B = \begin{bmatrix} I_x & 0 & 0 \\ 0 & I_y & 0 \\ 0 & 0 & I_z \end{bmatrix}. \quad (1)$$

The rotor angular velocities on the x_b and y_b axes have opposite signs ($w_{1,3} > 0$, $w_{2,4} < 0$) to counterbalance the reaction torque induced by the rotor and to control the yaw angle. The center of mass (COM) is located at $(0, 0, h_{COM})$

$$h_{COM} = \frac{(\frac{h_b}{2} \cdot m_b + (\frac{h_p}{2} + h_b) \cdot 4m_p)}{m}, \quad (2)$$

where h_b, m_b, h_p, m_p are the height and mass of the body and propellers respectively, and m the total mass of the quadrotor.

The position in the inertial frame is expressed by the vector ξ while the attitude, defined as Euler angles coordinates in the inertial frame, is represented by the vector η . Position and attitude are then grouped in the vector $q \in \mathbb{R}^6$.

$$\xi = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad \eta = \begin{bmatrix} \phi \\ \theta \\ \psi \end{bmatrix}, \quad q = \begin{bmatrix} \xi \\ \eta \end{bmatrix}. \quad (3)$$

In the body frame, the linear and the angular velocities are expressed by the vectors V_b and ν , respectively

$$V_B = \begin{bmatrix} v_{x,B} \\ v_{y,B} \\ v_{z,B} \end{bmatrix}, \quad \nu = \begin{bmatrix} p \\ q \\ r \end{bmatrix}. \quad (4)$$

The rotation matrix from the body frame to the inertial frame for this angle configuration is computed as the 3-2-1 Euler rotation matrix

$$R_{321} = \begin{bmatrix} c(\theta)c(\psi) & s(\phi)s(\theta)c(\psi) - c(\phi)s(\psi) \\ c(\theta)s(\psi) & s(\phi)s(\theta)s(\psi) + c(\phi)c(\psi) \\ -s(\theta) & s(\phi)c(\theta) \end{bmatrix} \begin{bmatrix} c(\phi)s(\theta)c(\psi) + s(\phi)s(\psi) \\ c(\phi)s(\theta)s(\psi) - s(\phi)c(\psi) \\ c(\phi)c(\theta) \end{bmatrix} \quad (5)$$

For concise notation define $s(x) = \sin(x)$, $c(x) = \cos(x)$, and $t(x) = \tan(x)$. The rotation from the inertial frame to the body frame is given by the transpose of R_{321} due to the orthogonality property of rotation matrices.

The transformation matrix W allows to invert the kinematics from the inertial frame to the body frame such that

$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & s(\phi)t(\theta) & c(\phi)t(\theta) \\ 0 & c(\phi) & -s(\phi) \\ 0 & s(\phi)/c(\theta) & c(\phi)/c(\theta) \end{bmatrix}}_{W^{-1}} \begin{bmatrix} p \\ q \\ r \end{bmatrix}, \quad (6)$$

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & -s(\theta) \\ 0 & c(\phi) & c(\theta)s(\phi) \\ 0 & -s(\phi) & c(\theta)c(\phi) \end{bmatrix}}_W \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix},$$

W is invertible if $\theta \neq (2k-1)\phi/2$, ($k \in \mathbb{Z}$). (i.e. W is not singular).

Each rotor produces a force T_i in the direction of the rotor z -axis and proportional to the square of the rotor velocity ω_i and to the lift constant k [15].

$$T_i = k\omega_i^2. \quad (7)$$

The rotor angular velocities and acceleration also produce the torque τ_{Ri} around the rotor axis with b and I_R being the drag constant and the inertia moment of the rotor respectively.

$$\tau_{Ri} = b\omega_i^2 + I_R\dot{\omega}_i. \quad (8)$$

For low speed maneuvers, it is common practice to consider k and b as constants despite their dependence on air density, propeller shape and angle of attack [10].

The total thrust along the body frame z -axis is obtained by combining each rotor's vertical force contributions

$$T_B = \begin{bmatrix} 0 \\ 0 \\ T \end{bmatrix}, \quad T = \sum_{i=1}^4 T_i. \quad (9)$$

The torque vector τ_B represent the torques in the direction of the respective body frame angles

$$\tau_B = \begin{bmatrix} \tau_\phi \\ \tau_\theta \\ \tau_\psi \end{bmatrix} = \begin{bmatrix} lk(\omega_4^2 - \omega_2^2) \\ lk(\omega_3^2 - \omega_1^2) \\ \tau_{R1} - \tau_{R2} + \tau_{R3} - \tau_{R4} \end{bmatrix}, \quad (10)$$

where l is the length of the quadrotor arm.

As a first approximation, the effect of $\dot{\omega}_i$ in τ_ψ is considered to be null.

A. Euler-Lagrange equations

The quadrotor dynamics may be described by the equation of motion expressed in the following form [13]:

$$B(q)\ddot{q} + C(q, \dot{q})\dot{q} + g = \zeta \quad (11)$$

where:

- $B(q) \in \mathbb{R}^{6 \times 6}$ is the inertia matrix
- $C(q, \dot{q}) \in \mathbb{R}^{6 \times 6}$ is the Christoffel matrix accounting for centrifugal and Coriolis effect
- $g \in \mathbb{R}^6$ is the vector accounting for the gravitational contribution
- $\zeta \in \mathbb{R}^6$ is the vector of forces in the inertial reference frame and torques acting on the UAV.

Equation (11) can be derived in a systematic way following Lagrangian formulation. The Lagrangian of a mechanical system is defined as the sum of translational and rotational kinetic energy $\mathcal{T}$ minus the potential energy $\mathcal{U}$

$$\mathcal{L} = \mathcal{T} - \mathcal{U}, \quad (12)$$

which leads to

$$\mathcal{L}(q, \dot{q}) = \frac{1}{2} m \dot{\xi}^T \dot{\xi} + \frac{1}{2} \dot{\eta}^T J \dot{\eta} - mg_z z, \quad (13)$$

where $J = W^T I_B W$ is the Jacobian matrix expressing the moment of inertia in the inertial reference frame, and g_z is the gravitational acceleration.

The Lagrange equation is given by

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}} \right)^T - \left(\frac{\partial \mathcal{L}}{\partial q} \right)^T = \zeta, \quad (14)$$

and sets the relationship between the generalized forces applied to the quadrotor ζ and the positions q , velocities $\dot{q}$ and accelerations $\ddot{q}$ [13], which is rewritten as:

$$\begin{bmatrix} f \\ \tau \end{bmatrix} = \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right)^T - \left(\frac{\partial \mathcal{L}}{\partial q_i} \right)^T, \quad (15)$$

$$i = 1, \dots, 6$$

where the external forces and torques acting on the UAV are:

$$f = RT_B, \quad \tau = \tau_B. \quad (16)$$

By solving (15), the equation of motion is derived which contains the following matrices:

$$B = \begin{bmatrix} M & 0 \\ 0 & J \end{bmatrix}, \quad C = \begin{bmatrix} 0_{3 \times 3} & 0 \\ 0 & C_\tau \end{bmatrix}, \quad (17)$$

$$g = \begin{bmatrix} 0 & 0 & mg_z & 0 & 0 & 0 \end{bmatrix}^T,$$

where $B, C \in \mathbb{R}^{6 \times 6}$, $M = m \cdot I_{3 \times 3}$ is the mass matrix, and $C_t(\eta, \dot{\eta}) = \dot{J} - \frac{1}{2} \frac{\partial}{\partial \eta_i} (\dot{\eta}^T J)$ is the matrix accounting for the centrifugal and Coriolis effect induced by the angular velocity.

To further improve model accuracy, the gyroscopic effect and the aerodynamic drag are considered and included in the model.

B. Gyroscopic Effect

The Gyroscopic effect is modeled as in [10]

$$G_{gyro} = \sum_{i=1}^4 S(\nu)_\times I_r \omega_i \mathbf{e}_3, \quad (18)$$

where $S(\nu)_\times$ is the skew symmetric matrix with ν components. Equation (18) can be re-written by factoring out the vector of position and Euler angle derivatives as

$$G_{gyro} = G_a(q, \dot{q}, \omega) \dot{q}, \quad (19)$$

where

$$G_a^T = \begin{bmatrix} 0_{3 \times 3} & 0 & 0 & 0 \\ 0 & 0 & -I_{r_z} \left(\sum_{i=1}^4 w_i \right) & 0 \\ 0 & I_{r_z} c_\phi \left(\sum_{i=1}^4 w_i \right) & 0 & 0 \\ 0 & I_{r_z} s_\phi \left(\sum_{i=1}^4 w_i \right) & I_{r_z} s_\theta \left(\sum_{i=1}^4 w_i \right) & 0 \end{bmatrix}^{6 \times 6}. \quad (20)$$

C. Aerodynamic drag

The aerodynamic forces acting on the quadrotor are modelled as [14], and expressed in the matrix form to adapt to the Lagrange formulation. The resulting drag forces account for multiple aerodynamic effects:

- Blade Flapping causes the rotor disk to tilt back when performing translational maneuvers. This effect opposes the motion of the UAV resulting in a drag force applied at the propeller COM. Considering a small advance ratio $\mu = \frac{|v_p|}{\varpi r}$, where V_p and ϖr are the horizontal velocity of the quadrotor and the rotor tip velocity respectively, The blade flapping drag is modelled as

$$D_{flap,i} = T_i \left(A_{flap} \frac{V_{pi}}{\omega_i} + B_{flap} \frac{\nu}{\omega_i} \right), \quad (21)$$

where

$$A_{flap} = \frac{1}{r} \begin{bmatrix} -A_{1c} & A_{1s} & 0 \\ -A_{1s} & -A_{1c} & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad (22)$$

and

$$B_{flap} = \begin{bmatrix} -B_2 & B_1 & 0 \\ B_1 & -B_2 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad (23)$$

are the parameter matrices to be identified empirically from testing.

- Induced Drag is due to the thrust imbalance on the rotor which causes the advancing blade to generate more lift than the retreating one. This effect occurs primarily in small scale quadrotor UAVs due to the relatively rigid rotor blades and opposes the motion in the direction of apparent wind. The induced drag is therefore modelled as a drag force applied at the propeller COM

$$D_{I,i} = K_I V_{pi}. \quad (24)$$

- Translational Drag is generated when the rotor moves in forward flight in the same way that airfoils experience induced drag associated with an increase in lift. The force model varies depending on the translational speed of the rotor

$$D_{T,i} = \begin{cases} K_{T_1} V_{pi} & V_{pi} < \bar{V}_{pi} \\ K_{T_2} (V_{z,i} - v_i)^4 V_{pi} & V_{pi} > \bar{V}_{pi} \end{cases}, \quad (25)$$

where $V_{z,i}$ is the rotor z -axis linear velocity expressed in the body frame, v_i is the vertical velocity induced through the rotor, and $\bar{V}_{pi}$ is the threshold velocity which depend on the propeller. This force is applied to the rotor COM and for simplicity the threshold velocity is not surpassed, therefore, only the first functional relationship is considered.

- Profile Drag is due to the transverse velocity on the propeller moving in to the air. This effect is not present during hovering but causes an increased drag when moving on body x, y -plane. The following linear model is a simplified variation of the complete nonlinear one derived from Blade Element Theory[14] and is applied at the propeller COM

$$D_{P,i} = K_P V_{pi}. \quad (26)$$

- Parasitic Drag is the resulting resistance force due to the flight envelope of the UAV. It's usually neglected for small scale quadrotor moving at speed lower than 10 m/s. The parasitic drag is modelled as

$$\begin{aligned} D_{par} &= K_{par} |V_B| V_B, \\ K_{par} &= \frac{1}{2} \rho S C_{D_{par}}, \end{aligned} \quad (27)$$

where ρ is the air density, S is the quadrotor lateral area and $C_{D_{par}}$ is drag coefficient. The resulting force is applied to the quadrotor COM.

The drag coefficients $K_I, K_{T_1}, K_{T_2}, K_P$ depend on the propeller shape and need to be empirically estimated.

The aerodynamic forces acting on the propeller COM are grouped in the vector

$$D_i = D_{flap,i} + D_{I,i} + D_{T_1,i} + D_{P,i}, \quad (28)$$

and the torques induced on the UAV is modelled as

$$\begin{aligned} \tau_{D_1} &= S(D_1)_{\times} (d\mathbf{e}_1 + h\mathbf{e}_3) \\ \tau_{D_2} &= S(D_2)_{\times} (-d\mathbf{e}_2 + h\mathbf{e}_3) \\ \tau_{D_3} &= S(D_3)_{\times} (-d\mathbf{e}_1 + h\mathbf{e}_3) \\ \tau_{D_4} &= S(D_4)_{\times} (d\mathbf{e}_2 + h\mathbf{e}_3). \end{aligned} \quad (29)$$

Finally the total aerodynamic forces and torques are expressed as

$$D_b = D_{par} + \sum_{i=1}^4 D_{flap,i} + D_{I,i} + D_{T_1,i} + D_{P,i}, \quad (30)$$

$$\tau_D = \sum_{i=1}^4 \tau_{D_i} \quad (31)$$

By factoring out the position and attitude derivative vector $\dot{q}$ the drag forces and torques are expressed in the following matrix form

$$\begin{bmatrix} D_b \\ \tau_D \end{bmatrix} = F_{flap}(q, \dot{q}, \omega) \dot{q} + F_I(q, \dot{q}) \dot{q} + F_T(q, \dot{q}) \dot{q} + F_P(q, \dot{q}) \dot{q} + F_{par}(q, \dot{q}, |\dot{q}|) \dot{q}, \quad (32)$$

$F_{flap}, F_I, F_T, F_P, F_{par} \in R^{6 \times 6}$. In the interest of space, the matrices components are not reported here.

D. Dynamic Equations of Motion

The gyroscopic effect and the aerodynamic forces are added to (11). Therefore, the complete model becomes

$$\begin{aligned} B(q) \ddot{q} + C(q, \dot{q}) \dot{q} + g + G_a(q, \dot{q}, \omega) \dot{q} + \\ + F_{flap}(q, \dot{q}, \omega) \dot{q} + F_I(q, \dot{q}) \dot{q} + F_T(q, \dot{q}) \dot{q} + \\ + F_P(q, \dot{q}) \dot{q} + F_{par}(q, \dot{q}, |\dot{q}|) \dot{q} = \zeta. \end{aligned} \quad (33)$$

The model may be simplified by grouping the aerodynamic effect matrices into a single one F

$$\begin{aligned} B(q) \ddot{q} + C(q, \dot{q}) \dot{q} + g + G_a(q, \dot{q}, \omega) \dot{q} + \\ + F(q, \dot{q}, |\dot{q}|, \omega) \dot{q} = \zeta(\omega). \end{aligned} \quad (34)$$

III. MODEL VALIDATION

The complete quadrotor model accounting for aerodynamic and gyroscopic effects, is simulated using Simulink. Fig. 2 shows the Simulink block diagram.

The model is validated by comparing it to a model developed using the Simscape toolbox, which allows for simulating multibody dynamics. (Fig. 3 shows the Simscape schematic). Simscape Multibody uses a dynamic engine to simulate the dynamical behaviour of 3D objects [16] and is able to achieve more realistic results by avoiding approximation errors.

The drag coefficients order of magnitude are chosen based on values found in literature [9] and then adjusted to resemble the "Relative Drag Variation with V" plot shown in [14]. The resulting drag-velocity plot is displayed in Fig. 4.

The Simulink and Simscape models are compared by feeding as input the same rotor velocities (Fig. 5) and measuring the error between the respective position and Euler angles.

Fig. 6 displays the Simscape quadrotor position and euler angles trajectory at the bottom, and the relative error between the Simulink and Simscape models at the top. Between the two models, the Simscape quadrotor is taken as the reference since it best approximate a real quadrotor system.

The resulting maximum error on position is around 0.18% while the one on the Euler angles is around 0.5%. This relatively low discrepancy between the two model shows the validity of the mathematical model implementation. Considering that the dynamic engine in Simscape Multibody was built based on the Newton Euler formulation [17], it would be expected that, using a model with such formulation, the error would be lower.

IV. THE CONTROL PROBLEM

Considering $\Omega \in \mathbb{R}^4$ as the input, which correspond to the square of the rotor angular velocity

$$\Omega = [\omega_1^2 \quad \omega_2^2 \quad \omega_3^2 \quad \omega_4^2]^T, \quad (35)$$

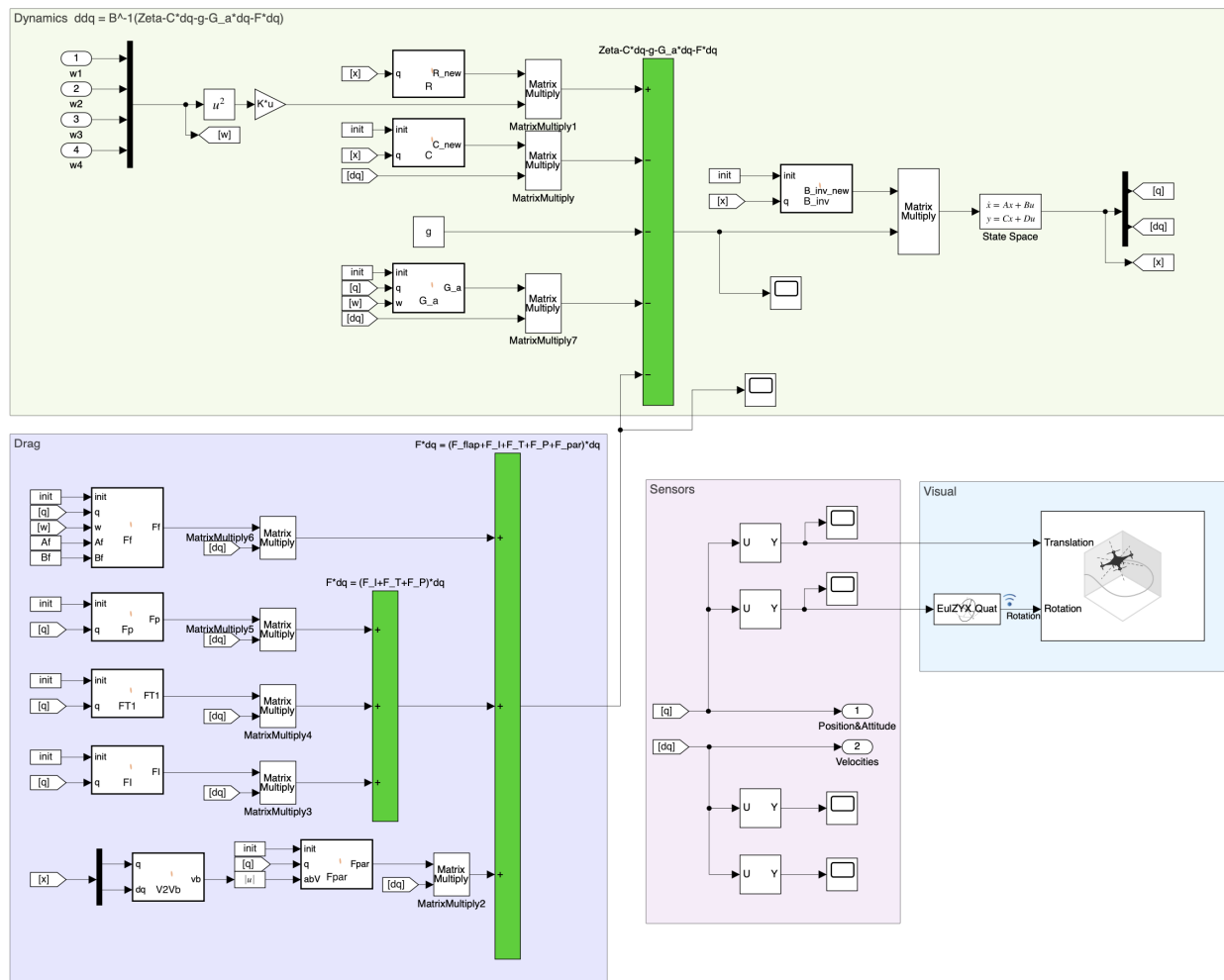


Fig. 2. Simulink Mathematical Model, iplementation of (34) in Matlab Simulink

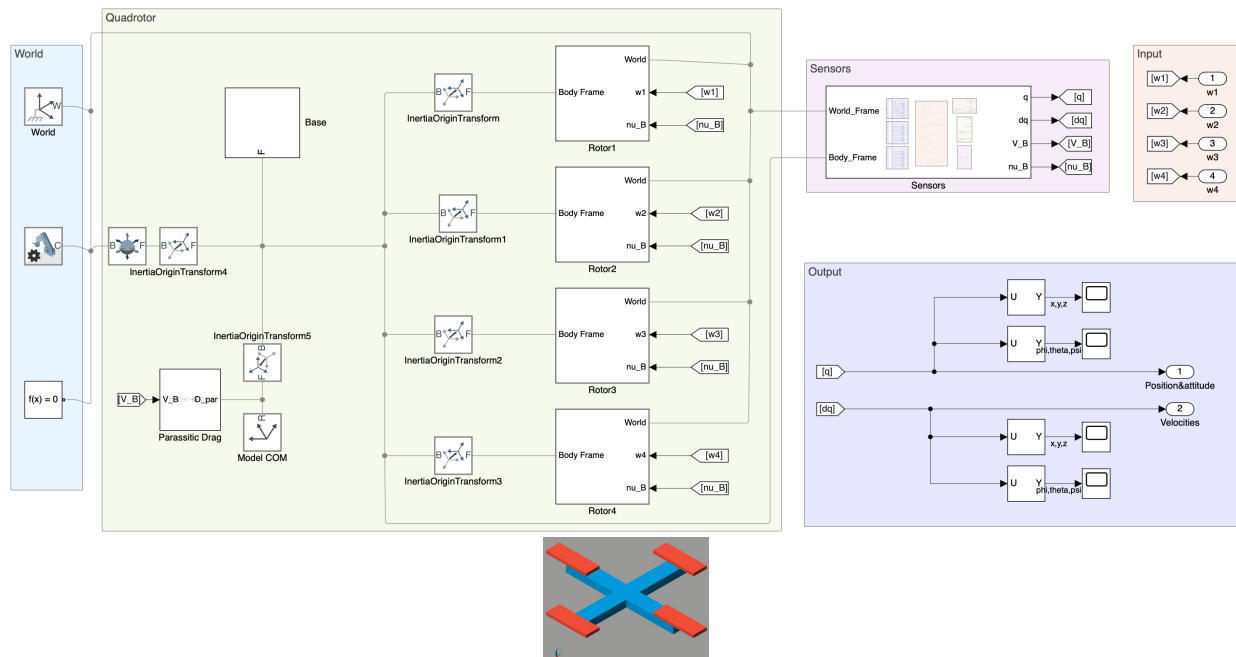


Fig. 3. Simulink Simscape Model and relative graphical representation

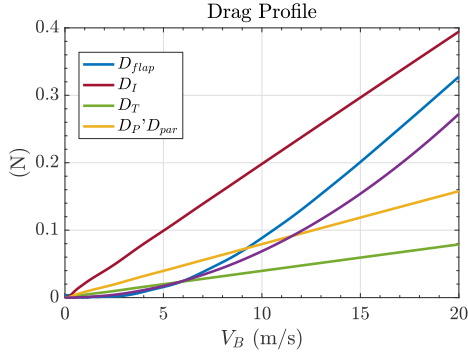


Fig. 4. Drag forces variation with V

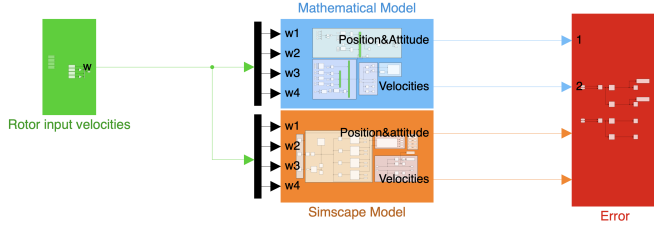


Fig. 5. Model Validation Setup, same rotor input velocities are fed to both simulink and simscape model and the response is compared

and (7), (9), (10), (16), the equation of motion (34) can be rewritten as

$$B\ddot{q} + C\dot{q} + g + G_a\dot{q} + F\dot{q} = \zeta = K_\zeta \Omega, \quad (36)$$

where $K_\zeta \in \mathbb{R}^{6 \times 4}$ is the matrix, derived from (7),(8), (9), (10), which establish the relation between the forces and torques and the angular velocities of the rotor

$$\begin{bmatrix} T_B \\ \tau_B \end{bmatrix} = K_\zeta' \Omega, \quad (37)$$

and since

$$f = R_{321}T_B, \quad \tau = \tau_B, \quad (38)$$

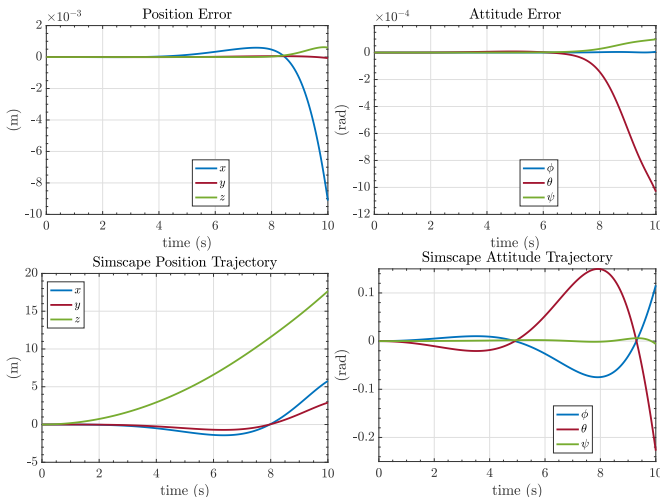


Fig. 6. Error between Simulink and Simscape model (at the top), Simscape quadrotor trajectory (at the bottom)

we get that

$$K_\zeta = \begin{bmatrix} R_{321} & 0 \\ 0 & I \end{bmatrix} K_\zeta'. \quad (39)$$

Since $\text{rank}(K_\zeta) = 4$, the quadrotor is an underactuated system. This can be intuitively checked, since the quadrotor has only 4 control inputs but six independent generalized coordinates [7]. As a consequence, the control inputs cannot accelerate the state of the UAV in arbitrary directions [18]. Therefore, a solution to this control problem can be found employing a hierarchical control approach which consist of a outer loop for position control and an inner loop for attitude tracking.

To further test the validity of the complete model, two controller implementation are simulated in this paper.

- PD controller with simplified model exact linearization
- PD controller with full model exact linearization

Both controllers are tested on the Simscape model in the Simulink environment.

A. Hierarchical PD trajectory control with simplified model feedback linearization

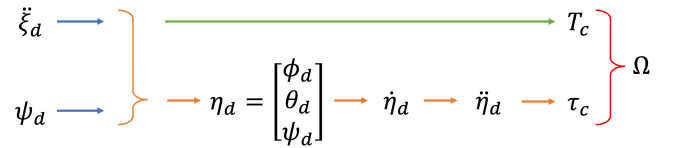


Fig. 7. Hierarchical control

Equations (7), (9), (10), define the dependencies between states, their derivatives and control inputs.

From Fig. 7 (from right to left) it can be seen that the square of rotor angular velocities Ω define the quadrotor total thrust T and torques τ . The torques influence angular accelerations hence, the angular velocities and Euler angles. On the other hand, the total thrust and Euler angles define the linear accelerations thus, the linear velocities and the linear positions. [15]. Therefore, to generate the proper control input Ω corresponding to a given position ξ or trajectory, these interactions need to be repeated backwards (from left to right of Fig. 7). The control problems is divided in two steps:

- The outer loop which is the position control loop
- The inner loop which is the attitude control loop

The outer loop control action is designed as in [10], where the position closed loop equation is

$$\ddot{\xi} = \ddot{\xi}_d + K_{P_p}(\xi_d - \xi) + K_{D_p}(\dot{\xi}_d - \dot{\xi}). \quad (40)$$

Considering the virtual control vector $U = \ddot{\xi} = (U_1, U_2, U_3)^T$, through (11), it follows

$$U = -g_z \mathbf{e}_3 + \frac{T}{m} R \mathbf{e}_3, \quad (41)$$

leading to

$$R^T(U + g_z \mathbf{e}_3) = \frac{T}{m} \mathbf{e}_3, \quad (42)$$

which, through some mathematical manipulation, like shown in [10], it allows to compute the desired pitch and roll attitude trajectory

$$\theta_{c1} = \arctan \left(\frac{U_1 \cos \psi + U_2 \sin \psi}{U_3 + g_z} \right), \quad (43)$$

$$\phi_{c1} = \arcsin \left(\frac{U_1 \sin \psi - U_2 \cos \psi}{\sqrt{U_1^2 + U_2^2 + (U_3 + g_z)^2}} \right), \quad (44)$$

which represent the attitude-commanded inputs of the inner loop. Equation (42) also allows to compute the total thrust generated by the four rotors as a function of the virtual control and the attitude

$$T_{c1} = m[U_1(\sin \theta \cos \psi \cos \phi + \sin \psi \sin \phi) + U_2(\sin \theta \sin \psi \cos \phi - \cos \psi \sin \phi) + (U_3 + g_z) \cos \theta \cos \phi] \quad (45)$$

The inner loop controls the torques τ with a PD controller enhanced by a feedback linearization. The control action is formulated as shown in [13]

$$\tau_{c1} = J(\eta)y + C_\tau(\eta, \dot{\eta})\dot{\eta}, \quad (46)$$

where the virtual control vector $y = \ddot{\eta}$ is

$$y = K_{Pa}(\eta_d - \eta) + K_{Da}(\dot{\eta}_d - \dot{\eta}). \quad (47)$$

After computing the desired thrust and torques, the controlled rotor squared velocities are given by

$$\begin{aligned} \begin{bmatrix} T \\ \tau_\phi \\ \tau_\theta \\ \tau_\psi \end{bmatrix} &= \begin{bmatrix} k(\omega_1^2 + \omega_2^2 + \omega_3^2 + \omega_4^2) \\ lk(\omega_4^2 - \omega_2^2) \\ lk(\omega_3^2 - \omega_1^2) \\ b(\omega_1^2 - \omega_2^2 + \omega_3^2 - \omega_4^2) \end{bmatrix} \\ &= \begin{bmatrix} k & k & k & k \\ 0 & -kl & 0 & kl \\ -kl & 0 & kl & 0 \\ b & -b & b & -b \end{bmatrix} \begin{bmatrix} \omega_1^2 \\ \omega_2^2 \\ \omega_3^2 \\ \omega_4^2 \end{bmatrix}. \end{aligned} \quad (48)$$

B. Hierarchical trajectory control with full model feedback linearization

The following control architecture is derived as in section above with the addition of gyroscopic and aerodynamic effect. To this end, (42) becomes

$$R^T(U + g_z \mathbf{e}_3 + \frac{1}{m}F(q, \dot{q}, |\dot{q}|, \omega)\dot{q}) = \frac{T}{m}\mathbf{e}_3, \quad (49)$$

which leads to the controlled roll and pitch angles

$$\theta_{c2} = \arctan \left(\frac{U_1^* \cos \psi + U_2^* \sin \psi}{U_3^* + g_z} \right), \quad (50)$$

$$\phi_{c2} = \arcsin \left(\frac{U_1^* \sin \psi - U_2^* \cos \psi}{\sqrt{U_1^{*2} + U_2^{*2} + (U_3^* + g_z)^2}} \right), \quad (51)$$

and the controlled thrust

$$T_{c2} = m[U_1^*(\sin \theta \cos \psi \cos \phi + \sin \psi \sin \phi) + U_2^*(\sin \theta \sin \psi \cos \phi - \cos \psi \sin \phi) + (U_3^* + g_z) \cos \theta \cos \phi] \quad (52)$$

where

$$U_i^* = U_i + F_i^*, \quad (53)$$

$$F_i^* = \frac{1}{m}\mathbf{e}_i^T F(q, \dot{q}, |\dot{q}|, \omega)\dot{q}. \quad (54)$$

Finally, the feedback linearization accounting for torque induced by the aerodynamic drag is

$$\tau_{c2} = J(\eta)y + C_\tau(\eta, \dot{\eta})\dot{\eta} + G_{a_\tau}(\eta, \dot{\eta}, \omega)\dot{\eta} + F_\tau(\eta, \dot{\eta}, |\dot{\eta}|, \omega)\dot{\eta} \quad (55)$$

The resulting control scheme consists of a cascade PD controller with feedback linearization.

V. SIMULATIONS RESULTS

The proposed controllers are simulated on the simscape model in the simulink environment. The quadrotor simscape plant has a sampling time of $1ms$ while the control input rate of is set to $10ms$ through a rate transitioning block. Fig. 8 displays the desired trajectory to be tracked by the quadrotor over a period of $20s$.

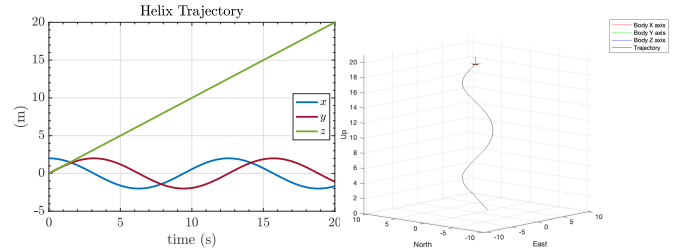


Fig. 8. Helix x, y, z -Trajectory

The parameters of the quadrotor are chosen based on the literature [6],[9] and reported in TABLE I, while the controller gains are derived experimentally in this work to ensure proper trajectory tracking.

Fig. 9 and Fig. 10 show the position and angular error followed by the normalized error of the two proposed controllers respectively.

Finally, the commanded rotor squared velocities of the controller without and with the aerodynamic drag compensation are shown in Fig. 12

From the figures it is noticeable that the two results are almost identical. This is to be expected since the aerodynamic effects are quite smaller than the quadrotor nonlinear dynamics. Besides, since the aerodynamic effect act on the quadrotor x, y -plane, they are orthogonal to the quadrotor z -axis and so, the linearization has no effect on the thrust computation [19]. However, taking a closer look at the steady state behaviour in Fig. 11, it can be seen that the controller with drag compensation is able to achieve, even if minimally, a smaller steady state error on the normalized position trajectory. The improved behavior is also verified

TABLE I
PARAMETER VALUES

Param.	Value	Name
m	1.04 kg	Total mass
l	0.2 m	Quadrotor arm length
h	0.168×10^{-1} m	Rotor z-offset from COM
I_x	7.691×10^{-3} kgm ²	Total moment of inertia along x_B
I_y	7.691×10^{-3} kgm ²	Total moment of inertia along y_B
I_z	15.254×10^{-2} kgm ²	Total moment of inertia along z_B
I_{rx}	1.406×10^{-5} kgm ²	Rotor moment of inertia along x_{rot}
I_{ry}	1.406×10^{-5} kgm ²	Rotor moment of inertia along y_{rot}
I_{rz}	2.813×10^{-5} kgm ²	Rotor moment of inertia along z_{rot}
k	1.149×10^{-5}	Thrust coefficient
b	2.690×10^{-8}	Rotor reaction drag coefficient
K_I	5×10^{-3} kgs ⁻¹	Induced drag coefficient
K_{T1}	1×10^{-3} kgs ⁻¹	Traslational drag coefficient
K_P	2×10^{-3} kgs ⁻¹	Profile drag coefficient
K_{par}	7×10^{-4} kgs ⁻¹	Parasitic drag coefficient
K_{P_p}	$5 \cdot I_{3 \times 3}$	Position proportional controller gain
K_{D_p}	$I_{3 \times 3}$	Position derivative controller gain
K_{P_a}	$90 \cdot \text{diag}(1, 1, 5)$	Attitude proportional controller gain
K_{D_a}	$9 \cdot \text{diag}(1, 1, 5)$	Attitude derivative controller gain
A_{1c}	20	A_{flap} Matrix component
A_{1s}	20	A_{flap} Matrix component
B_1	20	B_{flap} Matrix component
B_2	20	B_{flap} Matrix component

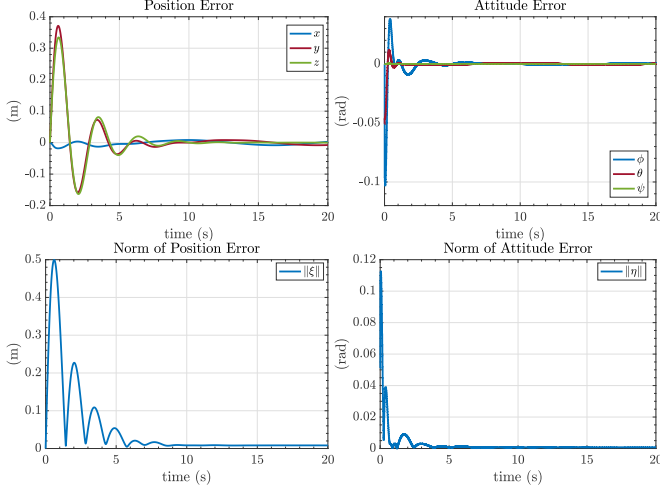


Fig. 9. Hierarchical PD trajectory control with simplified model feedback linearization: position and attitude error at the top, normalized position and attitude error at the bottom.

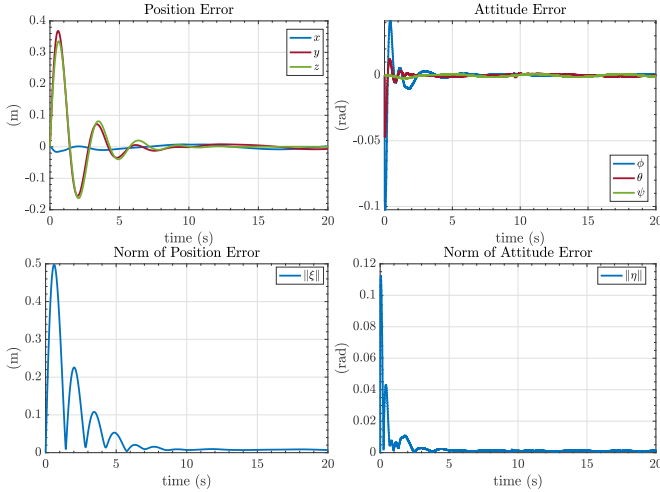


Fig. 10. Hierarchical PD trajectory control with full model feedback linearization: position and attitude error at the top, normalized position and attitude error at the bottom.

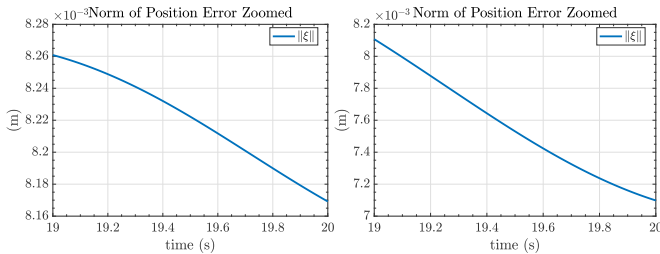


Fig. 11. Zoomed in steady state position error

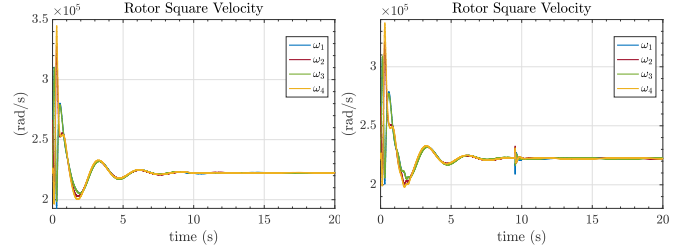


Fig. 12. Commanded Rotor Squared Angular Velocities: simplified model linearization on the left, full model linearization on the right

by comparing the root mean square error (RMSE) of the normalized position trajectory

$$RMSE_{\xi} = \sqrt{\frac{1}{20} \int_0^{20} \|e_{\xi}\|^2 dt}, \quad (56)$$

which for the simplified model compensation is $RMSE_{\xi_{simple}} = 0.1053m$ while for the full model linearization is $RMSE_{\xi_{full}} = 0.1047m$. Overall, using a fairly simple PD controller with feedback linearization, the quadrotor is able to perform and complete the desired helix trajectory with relatively low error.

VI. CONCLUSION

In this paper, a quadrotor model using the Lagrange formulation is explored. The model implements the aerodynamic forces presented in [14] and is simulated using Simulink environment. Furthermore, a multibody simulation using the Simscape toolbox is introduced, and allows for simulating the quadrotor as a 3D mechanical system using a dynamic engine to automatically solve the equations of motion. By providing the same rotor input velocity to both models, the

resulting maximum position and attitude error between the Simulink and Simscape models is below 0.5%, showing the fidelity of the mathematical model with respect to a more realistic simulation environment (Simscape model), even with the addition of aerodynamic effects. Moreover, a cascade PD hierarchical controller with exact linearization is then exploited for trajectory tracking problem. The feedback linearization is implemented first, using a simple model of the quadrotor, and then by compensating also for the gyroscopic effect and aerodynamic drag. The tracking results of the two controllers are comparable, with a slight improvement when using the complete model to linearize the dynamic. Therefore, the proposed model formulation allows for a compact matrix form which is relatively easy to derive and can be exploited for drag compensation in an effective control design.

As a future work, it could be advisable to test other controller designs and increase the aerodynamic coefficients to further test the improvements related to the aerodynamic compensation. Furthermore, adding some model uncertainties and disturbances, and simulating an equivalent model using ROS and Gazebo would test the robustness of the proposed controller in a more realistic simulation environment, in view of real hardware comparison. Moreover, further modelling refinement could be made by adding motor dynamics, ground effect [14] and ceiling effect [4].

Nevertheless, the result of this work will be used as a foundation for future work such as designing and testing quadrotor control applications. The authors intend to give free access on the quadrotor simulink and simscape models to be used as an intuitive test bed for control design and testing.

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